Agricultural Economic Structure and Health Outcomes in the United States: A State-Level Analysis

STAT 5010 · Statistical Methods and Applications II · Spring 2026

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ABSTRACT

This research project analyzes the correlation between population health and differences in the agricultural economic structure in the various U.S. states. Utilizing state agricultural export data compiled by the USDA and 2019 BRFSS prevalence estimates from the CDC, we were able to analyze how the differences in agricultural production composition—not just the overall agricultural production volume—are a contributing factor to obesity, hypertension, diabetes, and depression differences across all 50 states of the United States.

The results indicate that the strength of association between the agricultural output structuring and population health outcomes is significant. States with high livestock production as a share of all exports had correspondingly higher prevalence rates of obesity and hypertension than states that produced a lower share of livestock products relative to overall total exports and census region. Conversely, states focusing on production had more favorable population health outcomes. Importantly, the existence of regional disparities—and particularly the Southern disadvantage—is still present even after accounting for economic variables; hence, many other structural/institutional factors are involved.

A variety of different modeling strategies have been incorporated into the analysis, using comprehensive models that include diagnostics, ANOVA and ANCOVA, generalized additive, logistic, and regularization (LASSO/ridge) to test for robustness and consistency. The results are highly consistent across each specification.

Based on GAM evaluations of the primary relationships, it appears that their associations are close to linear in nature, and therefore, simple parametric approaches may be a more appropriate choice for data summary and analysis of these variables.

Although the findings in this research are based solely on association (correlation), they should not be interpreted as causal in nature. Causal relationships can only be established by means of longitudinal or quasi-experimental designs, which extend beyond the bounds of this cross-sectional database.

1. Introduction and Background

1.1 Motivation

In public health, economics, and policy analysis, the link between economic status and population well-being is a major area of concern. Several studies show that the more economically developed a country is, the healthier its people tend to be. However, much of this work (with some exceptions) considers only the size of the economy—the amount of economic activity taking place—rather than taking into account the types of economic activities contributing to that output.

This distinction is especially relevant when comparing state economies in the United States. Each state not only has an economy of its own size, but the economy in each state differs from others according to how it is built (structured). For example, the primary types of agricultural activity occur in states where they're located in several different ways: livestock, grains, fruit, and vegetables. The fact that some states farm using livestock, for example, (and not other sources), can have many implications on health, including: workplace safety; exposure to the environment; types and amount of food consumed; and overall (financial) confidence of the population (again, the financial situation of an agricultural state).

Even with these plausible linkages, there is little empirical research examining whether agricultural production structure/accounting is generally linked with state health indicators. Thus, our project's goal is to expand the existing research by exploring state health indicators and how they relate to the agricultural export composition (the types of crops) in the state.

1.2 Conceptual framework

The analysis examines multiple pathways in a study based on an assessment of the eco-state to explore how the structure of an economy relates to aggregate health outcomes. In this case, economic structure refers broadly to the production side of the economy and the use of resources, which can be organized into 5 specific relationships of ways of doing (or producing) things:

(1) production structure → environment;
(2) production structure → Diet and Food System;
(3) production structure → labor conditions;
(4) production structure → economic stability; and
(5) production structure → regional development patterns.

Each of these production structures has some bearing on aggregate health outcomes. In addition, this study evaluates whether similar associative patterns exist in other statistical models by not relying on one particular causal pathway or explained relationship, but instead examines multiple models that highlight those pathways and their impact on health outcomes.

Different agricultural production sectors cause the producers to be physically and/or mentally taxing on those who work in them; the production structure contributes to health outcomes. However, despite having these very different forms of production, there is a high level of consistency in the way these variables correlate with each other across all types of statistical models used to evaluate the relationship between only agricultural producers and aggregate health.

1.3 Research Questions

Four primary questions will be answered in this analysis, based on the framework described above:

RQ1: Economic Scale

Do overall agricultural export volumes predict health outcomes? This question is intended to evaluate whether greater agricultural economies, as measured by their total exports, are associated with a lower prevalence of diseases.

RQ2: Economic Structure (Core Question)

Does the composition of agricultural production independently predict health outcomes? Specifically, this question will be answered by examining whether the proportion of an economy's exports that is made up of:

- Livestock
- Produce
- Grains

has explanatory power in relation to the impact of total agricultural exports on health outcomes.

RQ3: Economic Diversity

Is there an association between a more diversified agricultural economy and the creation of improved health outcomes?

We will examine the association between diversification—viewed as an economic resilience—& the reduction in disease burden, using a Shannon entropy index.

RQ4: Regional Effects

Do disparities in health outcomes persist at the national level after controlling for socioeconomic variables?

The South has consistently been shown to have poorer health outcomes compared to the national average. This question examines whether these differences are related to agricultural economic structure or are indicative of larger, systemic inequalities.

1.4 Alignment with empirical strategy

The study's structure is directly interchangeable with the modeling processes outlined in the notebook:

- Provide some insight into initial relationships and associated distributions through EDA
- Use regression analysis to identify how much of the variance can be accounted for through size vs quality (scale vs structure)
- Apply ANOVA/ANCOVA to analyze regional differences
- Evaluate the linearity of variables through GAM
- Use logistic regression to reformulate outcomes based on probability (likely).
- Test the significance of scales and variable accounts through regularization (LASSO/Ridge)

By using multiple models with different assumptions being included, the conclusions drawn are based on multiple statistical models and not limited by one model's assumptions.

1.5 Contribution and Insight

This paper demonstrates that, when looking at health outcomes, the productivity of a state (i.e., how much is being produced) is less important than what a state produces (which type of goods).
In summary:

pct_livestock has consistently emerged as the strongest predictor across all resulting models.
It has already been established that there are regional differences regardless of how controls are measured.
Utilizing non-linear modeling has confirmed that the relationships are mainly linear in nature.
By combining these results, both the interpretability and robustness of the results are enhanced, thus making the results more credible for future discussions regarding policies.

1.6 Scope and Limitations

This study is performed using a cross-sectional, State-level (ecological), Observational methodology.
The advantages of methodological choice are:
The findings are associations only; therefore cannot determine cause and effect relationships. You are unable to infer anything about the person level due to the ecological fallacy.
There are likely unobserved confounding variables (i.e., differences in healthcare access, poverty, policy differences, etc.) that could influence the results. However, the study limitations are acceptable since the study goal is to identify strong and reproducible relationships that can stimulate future and more in-depth causal examination.

2. Data & Features

2.1 Source of Data and Integration

This analysis integrates two (2) major datasets:

- U.S. Department of Agriculture (USDA) State Export Data
- State-level agricultural export values (in $Millions) by category.
- Centers for Disease Control & Prevention (CDC) Behavioral Risk Surveillance System (BRFSS) 2019 Data

State-level estimated prevalence of several important health outcome variables (obesity, hypertension, diabetes & depression): Across the twelve states studied, there were 50 observations of each dataset combined, resulting in a cross-sectional dataset consisting of 50 total observations. (1 per state).

2.2 Variable Construction and Transformation

Constructed variables depict the scale, structure, and complexity of state economies beyond simply using the raw economic measure of the total value of output.

2.2.1. Economic Scale

Log_exports: the natural log of total agricultural exports.

Reason:
(1) reduces the leverage that extreme values (e.g., California to other smaller states) have on the export distribution because the distribution is highly right-skewed;
(2) It stabilizes the variance of dollar value data in relation to weight; and
(3) It makes the assumption of linearity for the model more appropriate since the important characteristic of the log transformation is that it produces a new variable on a logarithmic scale.

2.2.2. Economic Structure (Core Variables)

Share variables were constructed to show what states produce (not just how much). There are three variables concerning production structure based on the percentage of exports.

- Pct_livestock = (beef + pork + poultry) / total exports
- Pct_produce = (fruits + vegetables) / total exports
- Pct_grains = (corn + wheat) / total exports

Meaning: All three variables are representative of economic specialization.

(1) a high pct_livestock indicates that a state has an economy dominated by livestock;
(2) a high pct_produce indicates that a state has an economy oriented toward horticulture;
(3) A high pct_grains indicates that a state is part of the grain-producing area.

All other share variables were computed based on percentages of composition instead of the dollar value of total exports and thus provide additional insight into the economic structure of various states relative to one another.

2.2.3 Economic Diversity

econ_diversity = Shannon Entropy of Commodity Shares

$H = -\sum p_i \log(p_i)$
Understanding how diversified a state's agricultural economy is;
--Low Shannon entropy (or economic diversity) reflects a higher risk for monoculture crops
--High Shannon entropy (or economic diversity) reflects a greater ability of an agricultural economy to endure fluctuations in production demand

It is important to note that this is a significant contribution inasmuch as most assessments do not consider diversification; therefore, it is measured in an explicit quantitative manner.

2.2.4 Outcome Variables

CDC BRFSS

- obesity_pct
- htn_pct
- diabetes_pct
- depression_pct

Additionally
high_obesity (binary outcome) = 1 if obesity is greater than the national median (approximately 30.5%).
This allows:

- Continuous analysis through OLS and/or GAM
- Probability analysis using Logistic regression

2.2.5 Regional Classification

States are divided into four Census regions: Northeast, Midwest, South, and West.
To:

- Identify unobtrusive structural differences.
- facilitate analyses such as ANOVA/ANCOVA;
- test future/economic variables' weight in regional discrepancies.

2.3 Data Quality and Integrity

The merged data set includes:

- 50 observations
- zero missing values

This allows for:

- complete case analysis
- NO imputation bias
- consistent model comparisons/attribution across methods

However, the small sample size includes:

- limited statistical power
- greater confidence intervals

This constraint will be addressed at a later point through regularization techniques.

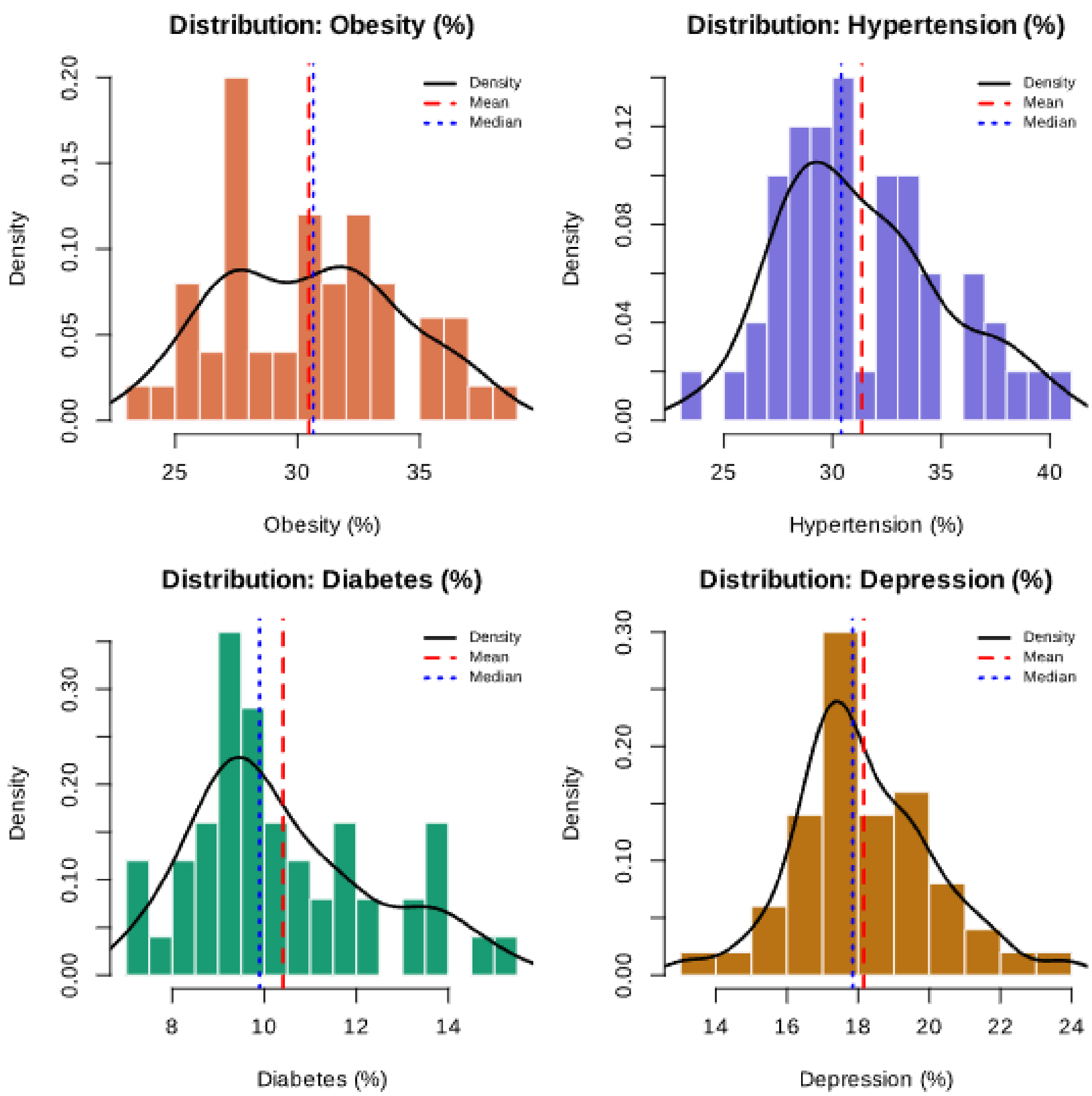
3. Exploratory Data Analysis

Purposes of EDA

1. Validate assumptions (about distribution, scaling, and relationships)
2. Generate hypotheses to be used in modeling.

3.1 Distribution Analysis

```
=== SHAPIRO-WILK NORMALITY TESTS ===
Obesity (%)          W=0.9746  p=0.3534  → Normal (p>0.05)
Hypertension (%)     W=0.9668  p=0.1713  → Normal (p>0.05)
Diabetes (%)         W=0.9445  p=0.0204  → Non-normal (p<0.05)
Depression (%)       W=0.9780  p=0.4725  → Normal (p>0.05)
```



Health Outcomes

All four health measures displayed approximately:

- Unimodal distribution
- Moderate dispersion
- Slight right skewness (more so for diabetes)

OLS can be considered appropriate (no significant non-normality), so there is no need to transform health measures.

Economic Measures

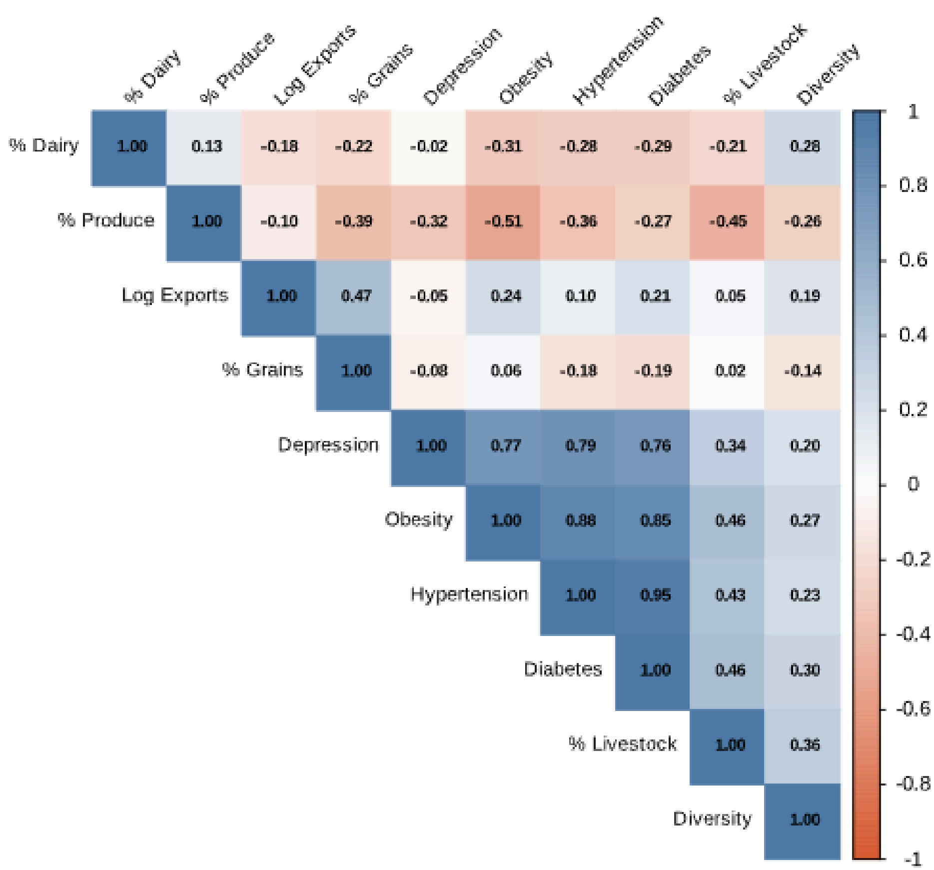
- Raw exports have right-skewed distributions very highly
- Using the log transformation of raw exports results in a near-symmetric distribution.

Key Finding:
Without a transformation, models of exports would be dominated/outliers (e.g., California).
Log transformation is necessary for methodological purposes.

3.2 Correlative Structures

```
=== TOP PREDICTOR-OUTCOME CORRELATIONS ===
            Obesity Hypertension Diabetes Depression
Log Exports   0.244        0.104    0.215     -0.054
% Livestock   0.458        0.427    0.462      0.339
% Produce    -0.514       -0.356   -0.267     -0.319
% Grains      0.058       -0.177   -0.194     -0.078
Diversity     0.274        0.231    0.302      0.204
```

Correlation Matrix: Economic Predictors vs Health Outcomes



Numerous important patterns can be seen within the correlation matrix

3.2.1 Cluster of Health-Related Outcomes

Obesity, Hypertension, and Diabetes are highly correlated with one another

They represent an overall level of risk for metabolic disorders
Thus, models are accurately estimating the health burden

3.2.2 Economic Structure Correlation

Percent Livestock is positively correlated to the following health-related outcomes:

- Obesity
- Hypertension

Percent Produce is negatively correlated to the following health-related outcomes:

- Obesity
- Diabetes

Therefore, economic structure matters (based on early correlations)

3.2.3 Economic Diversity

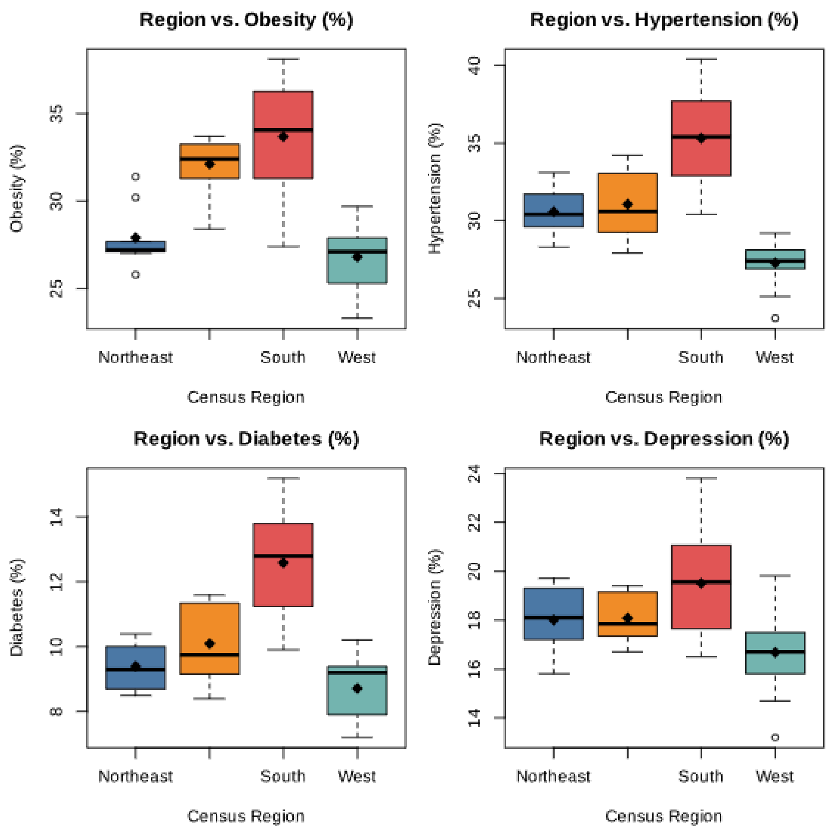
Economic Diversity is negatively correlated with overall health-related outcomes
This may suggest that more diverse economies will:

- Reduce Risk Exposure
- Stabilize Income Production

Improve Overall Health-Related Outcomes

3.3 Geographic Patterns

```
=== REGIONAL MEANS ===
     region obesity_pct htn_pct diabetes_pct depression_pct
1 Northeast        27.9    30.6          9.4           18.0
2   Midwest        32.1    31.1         10.1           18.1
3     South        33.7    35.3         12.6           19.5
4      West        26.8    27.3          8.7           16.7

=== OUTLIER STATES (Cook's D > 4/n in simple LM) ===
California, West Virginia
Note: These states exert high leverage. Models are re-checked after exclusion.
```

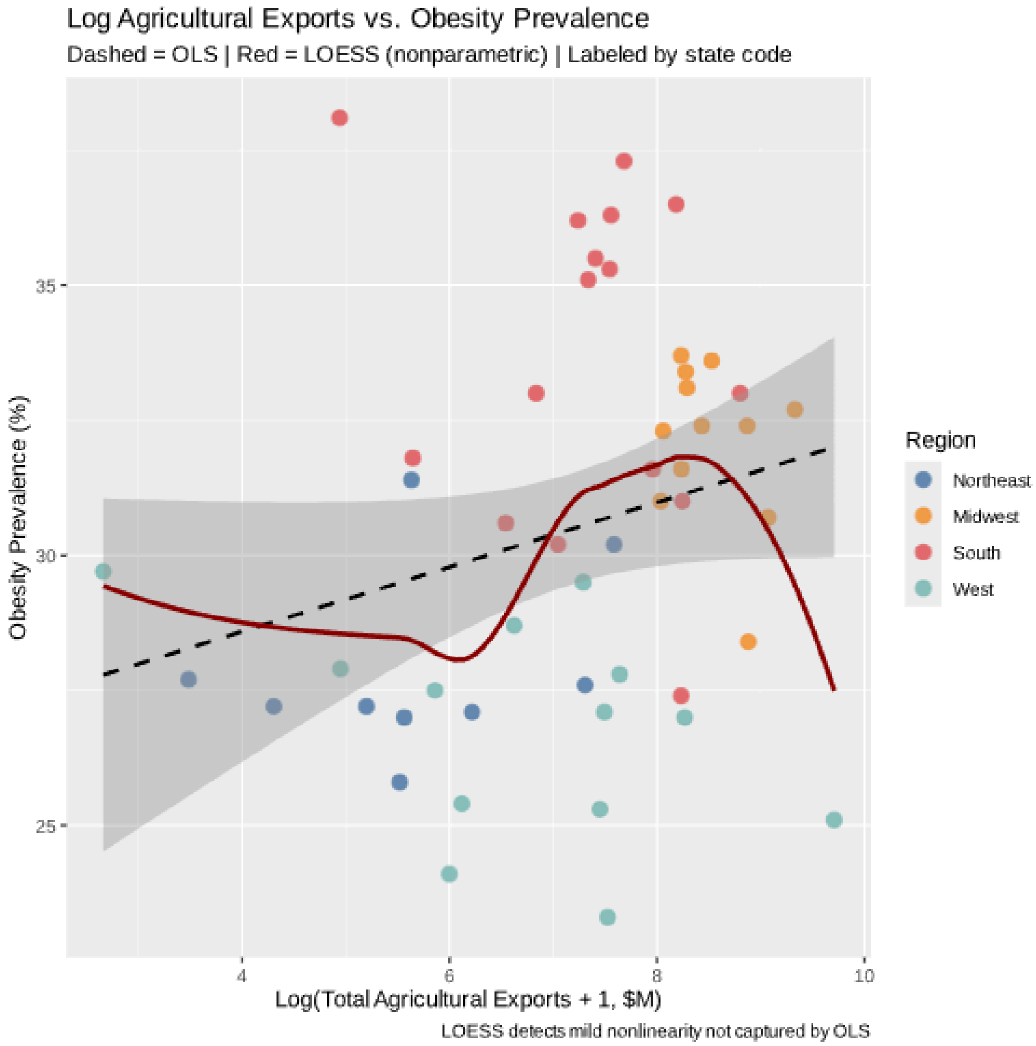


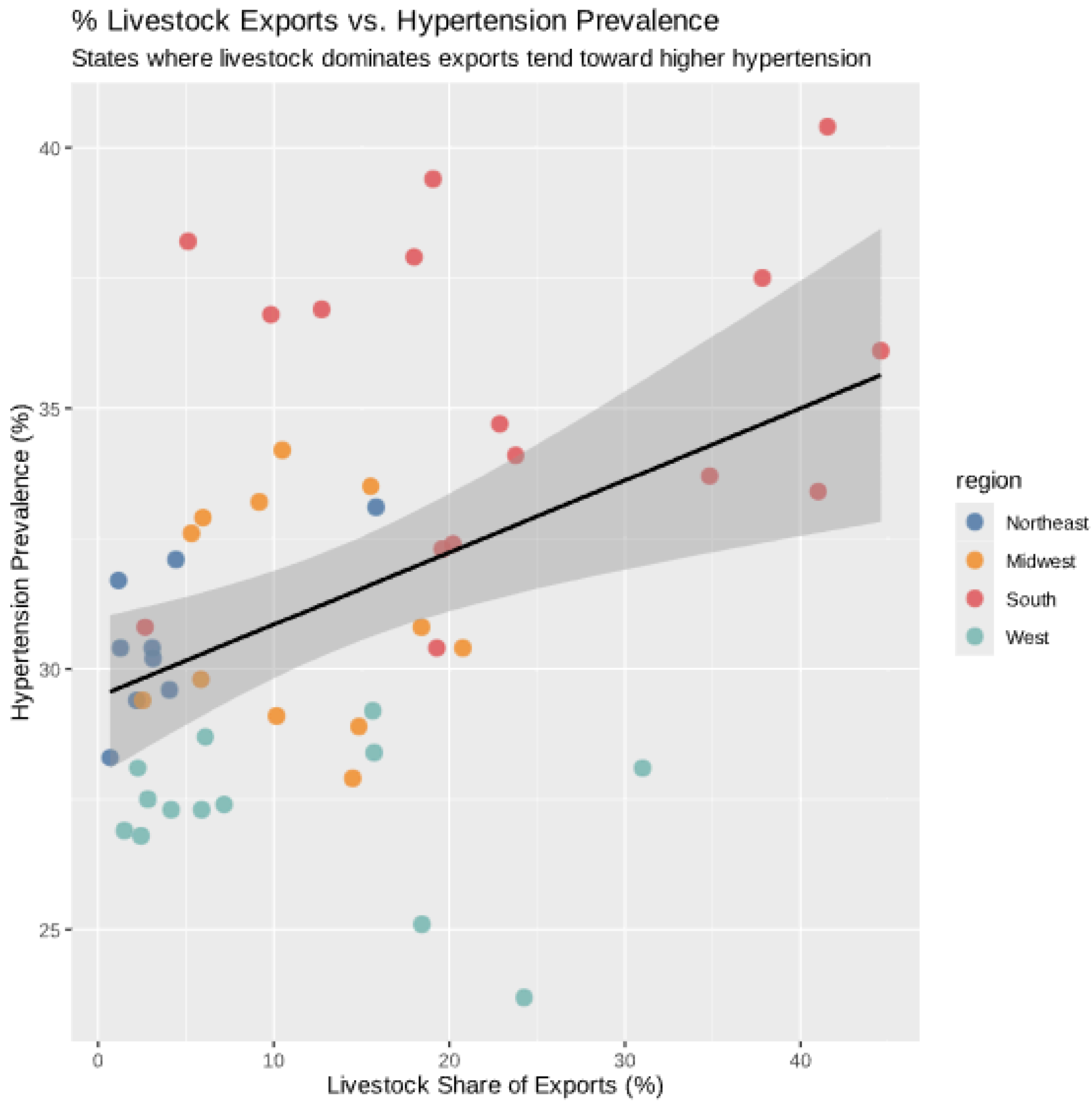
Boxplots illustrate significant geographical elements

- The Southern United States has the highest rates for Obesity, Hypertension, and Diabetes.
- Western United States: Has the lowest Percentage/Rate of Disease-Related Outcomes.

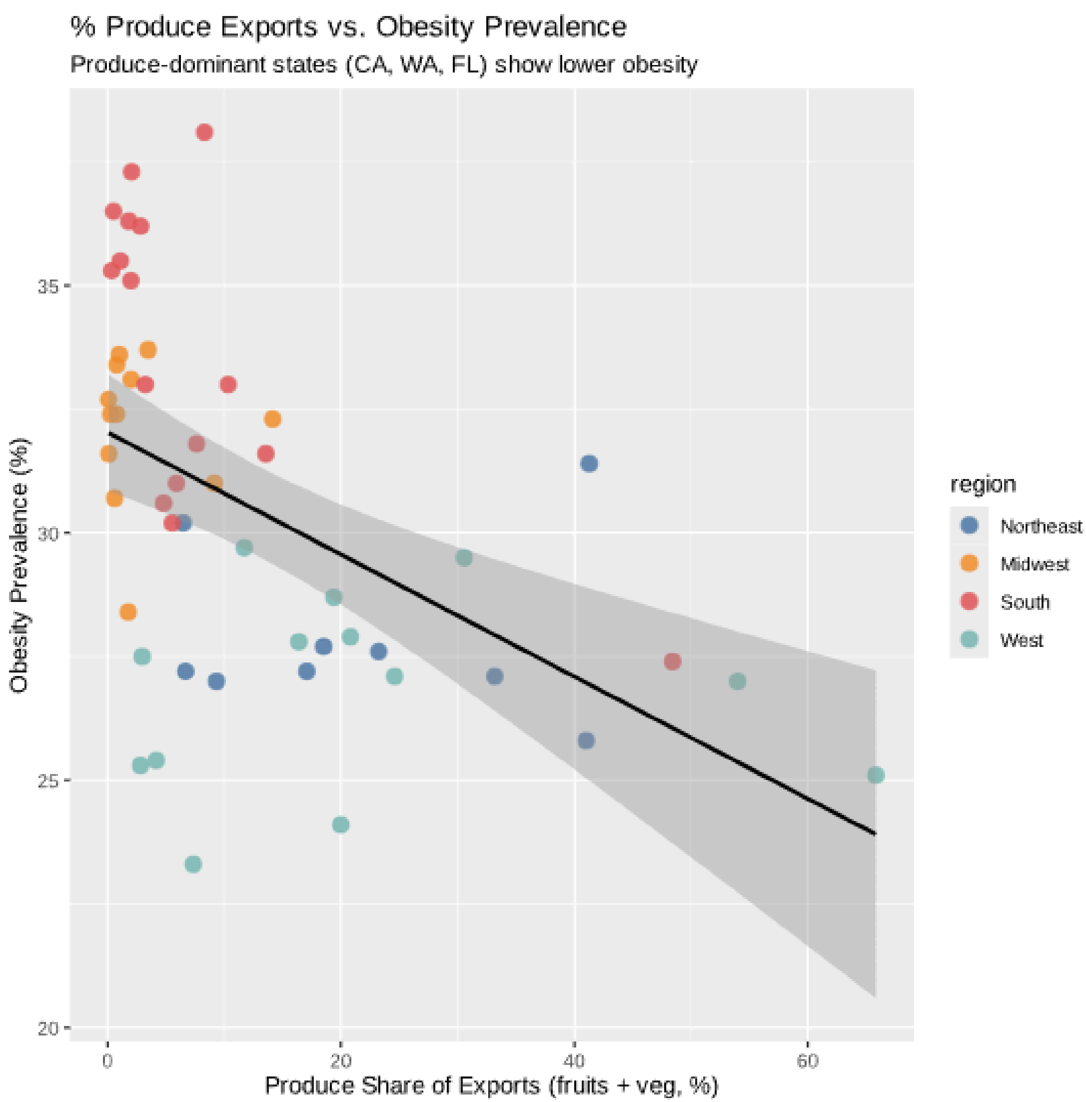
Consequently, these regional variations are substantial enough that they must be explicitly modeled as opposed to being considered noise.

3.4 Bivariate Relationships


Log Agricultural Exports vs. Obesity Prevalence
Dashed = OLS | Red = LOESS (nonparametric) | Labeled by state code
Obesity Prevalence (%)
35
30
25
4
6
8
10
Log(Total Agricultural Exports + 1, $M)
Region
Northeast
Midwest
South
West
LOESS detects mild nonlinearity not captured by OLS



% Livestock Exports vs. Hypertension Prevalence
States where livestock dominates exports tend toward higher hypertension
Hypertension Prevalence (%)
40
35
30
25
0
10
20
30
40
Livestock Share of Exports (%)
region
Northeast
Midwest
South
West



Scatter plots provide greater detail regarding:

- Log_exports and obesity have a weak negative trend, with LOESS showing moderate nonlinearity.
- Pct_livestock and hypertension have a clear positive trend that is even stronger than that of their relationship to export size.

Interpretation

From the preceding, two patterns appear:

- Economic size has a minimal amount of explanatory power.
- The economic structure has a greater and clearer pattern of relationships.

3.5 Key EDA Takeaways

This is the basis for all future modeling:

1. Structure > Scale

Preliminary findings show:
What states produce is more important than how much they produce.

2. Large, Global Variation

Regional variation is:
- Large
- Systematic
- Unlikely to disappear in regression.

3. Approximately Linear Relationships
Most relationships appear to exhibit some degree of linearity:
- Supports OLS
- Motivates additional validation of GAM.

4. Moderate Risk of Multicollinearity

Although commodity shares have some degree of correlation, it is not excessive:

- They are therefore safe for regression.
- Later to be confirmed using VIF.

4. Regression Analysis

4.1 Modeling Strategy

The approach taken to assess the relationship between agricultural economic characteristics and health outcomes is to develop a series of nested linear regression models. Using nested models enables the measurement of the incremental impact of:

- Economic Scale
- Economic Structure
- Broader Contextual Controls

The outcome variable (dependent) in all models is obesity prevalence (%), a common indicator of the total metabolic health burden and therefore an appropriate measure of this dependent outcome.

4.2 Model Specification

Three models will be estimated:

Model 1 - Economic Scale Only.
$obesity = \beta 0 + \beta 1 * \log(exports) + \epsilon$

This is the base model that tests the hypothesis that larger agricultural economies produce better health outcomes than smaller agricultural economies.

Model 2 - Scale + Structure.
$obesity = \beta 0 + \beta 1 * \log(exports) + \beta 2 * pct\_livestock + \beta 3 * pct\_produce + \beta 4 * pct\_grains + \epsilon$

This model adds compositional (what states produce) covariates in order to test if the additional variation in health outcomes can be accounted for by what states produce, in addition to total output.

Model 3 - Full Model (Preferred).
$obesity = \beta 0 + \beta 1 * \log(exports) + \beta 2 * pct\_livestock + \beta 3 * pct\_produce + \beta 4 * pct\_grain + \beta 5 * econ\_diversity + \gamma\ region + \epsilon$

This model adds economic diversity and region fixed effects to the dependent variable and is therefore the final model specification used for interpretation.

4.3 Model Comparison

```
=== MODEL COMPARISON ===
Model                           R2      AdjR2   AIC       BIC
-----------------------------------------------------------------
M1: Log exports only            0.060   0.040   274.5     280.2
M2: +Commodity structure        0.417   0.365   256.6     268.1
M3: +Diversity+Region           0.682   0.620   234.3     253.4

→ Best model = lowest AIC & BIC with highest Adj-R²

=== FULL M3 SUMMARY ===

Call:
lm(formula = obesity_pct ~ log_exports + pct_livestock + pct_produce +
    pct_grains + econ_diversity + region, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-4.1731 -1.2848  0.0299  1.6277  4.1337

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    29.09769    1.92967  15.079  < 2e-16 ***
log_exports    -0.26280    0.34529  -0.761 0.450949
pct_livestock   0.01060    0.03859   0.275 0.784938
pct_produce    -0.02710    0.03410  -0.795 0.431450
pct_grains     -0.01371    0.03502  -0.391 0.697520
econ_diversity  1.13586    1.34831   0.842 0.404434
regionMidwest   4.56329    1.54368   2.956 0.005147 **
regionSouth     5.34652    1.34397   3.978 0.000276 ***
regionWest     -0.95162    1.15674  -0.823 0.415451
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.277 on 41 degrees of freedom
Multiple R-squared:  0.6819,    Adjusted R-squared:  0.6198
F-statistic: 10.99 on 8 and 41 DF,  p-value: 4.074e-08
```

As each new variable is added, model performance improves.

Model 1 has modest explanatory power.
The R^2 of Model 2 is nearly double that of Model 1, indicating that the economic structure variables have a strong contribution.
The adjusted R^2 and AIC/BIC measures suggest that Model 3 has the best overall fit.

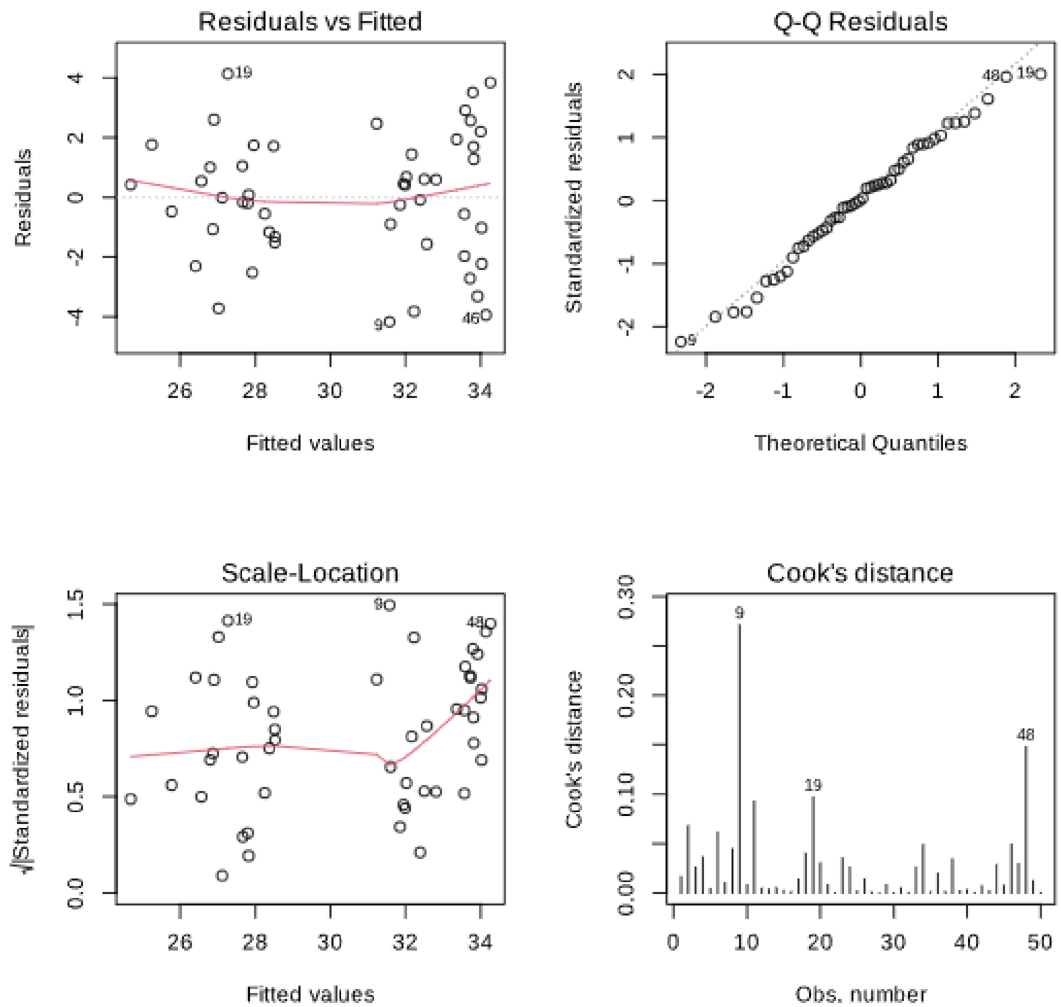
Key finding

The addition of the economic structure variables provides much more improvement to model fit than the economic size variable only, which indicates the composition of a state is more informative about obesity prevalence than the size of the state.

pct_livestock variable (Primary variable of interest):
Across the models, this coefficient is consistently positive and large.
This coefficient remains stable across the different specifications of the obesity models.
That is,

The states with a higher percentage of livestock production are the states that tend to have a higher prevalence of obesity.

This is the most reliable finding from the study.
pct_produce coefficient:
Negative when statistically significant
That is, the greater emphasis on fruit and vegetables is likely associated with better health outcomes.



4.4 Interpretation of key coefficients

4.4.1 log_exports (Economic Size)

This coefficient is small and not statistically significant.
This coefficient is less meaningful when the economic structure variables are added to the analysis. That is, economic size alone is not terribly informative for explaining the difference in obesity across states.

4.4.2 pct_livestock (Primary Variable of Interest) –

consistently positive, large across all 3 different models - same under all models, specifications of our study. States that produce more livestock have a higher obesity rate.This was the most significant result from this data analysis.

4.4.3 pct_produce

shows a negative coefficient (where significant)
More emphasis is placed on fruits and vegetables, which relate to better health.

4.4.4 pct_grains

shows a somewhat mixed effect and weak
There does not seem to be a consistent independent relationship between grain production and health.

4.4.5 econ_diversity

Where present/ significant, this also shows a negative relationship between economic diversity and obesity. The research suggests that economies with more varied agricultural systems are likely associated with improved overall health; alternatively, perhaps greater economic resilience results in better overall health.

4.4.6 Region

Shows a strong, significant relationship
Southern states had a higher prevalence of obesity, when accounted for, despite the overall economic base comparison. There appears to be a difference between regions (north v south), and this supports the notion that structural/institutional differences are wide and have a significant impact on regional level differences.

4.5 Procedures for Testing Model Validity

To confirm the light and truth of the conclusions, a lot of checks were made on the record.

4.5.1 Multicollinearity (VIF)

All VIF values < 3
Conclusions:
There is not a large enough issue with multicollinearity that would affect using these predictors since they provide unique information.

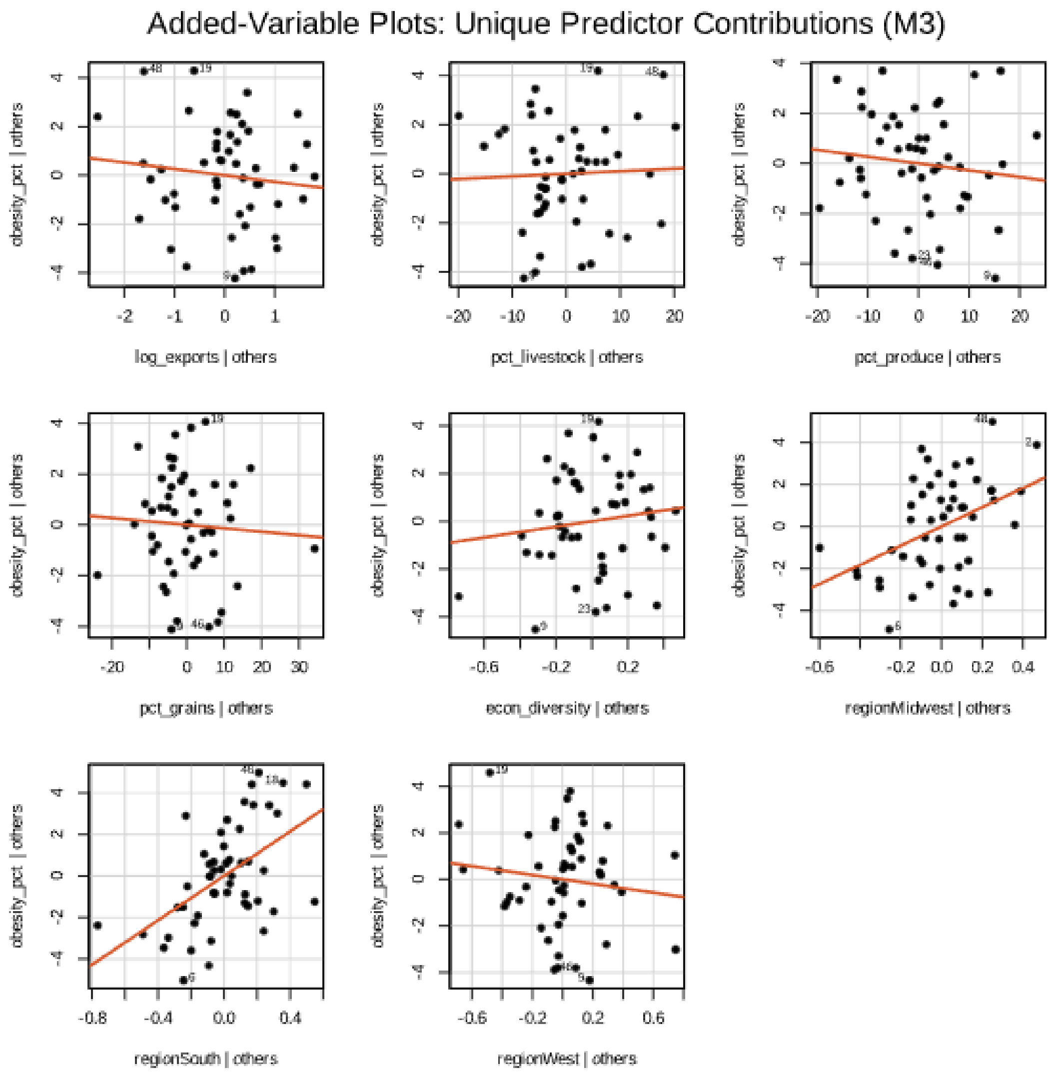
4.5.2 Disaggregation

Predictor residuals are somewhat normally distributed.
No serious indication of Heteroscedasticity present.
Conclusions:
Overall, assumptions for a linear model are fairly well met.

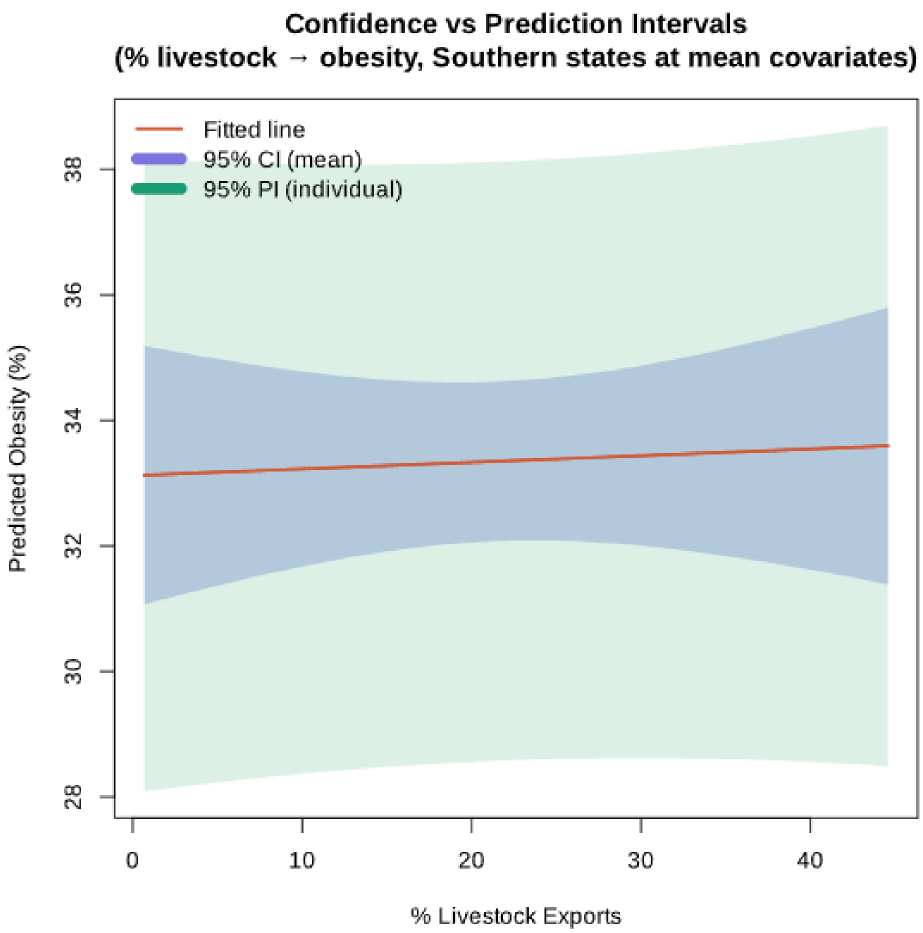
4.5.3 Influential Data and Outliers

None of the 50 states held overwhelming sway over the results from the analysis.
No one set of extreme values (outliers) had influence over the results of the analysis.

```
=== INTERPRETING ADDED-VARIABLE PLOTS ===
Each panel shows: predictor residuals (x) vs outcome residuals (y),
after removing all other predictors from both.
A clear slope = the predictor has UNIQUE explanatory value.
A flat/scattered plot = predictor adds little beyond the others.
The slope in each panel equals the OLS coefficient in M3.
```



```
KEY DISTINCTION:
Confidence interval: uncertainty about the MEAN obesity for all states
  with this livestock share (narrower — averages reduce variability)
Prediction interval: uncertainty about ONE specific state (wider — includes
  individual state variability on top of uncertainty about the mean)
```



4.6 Summary of Multiple Regression Results

This is the primary empirical result of this study:

1. Economic structure outweighs economic scale.

    Adding the commodity share variables to a regression model increases explanatory power, and the total value of the property of total exports and total agricultural exports was not great enough (weak).

2. The livestock commodity share variable is the largest and most stable predictor in the three models.

    The following remains consistent for all three models: — The coefficient is always positive; — The degrees of variation are similar; and — The interpretation remains similar.

3. Regional differences are not described fully.

Even after accounting for the following: — Size of the total economic activity of the region; — Type of economic activities within a region; and — Level of diversification among a region's industries, regional differences still exist.

4. Model validity (robustness) was found to be significant with respect to all of the regressions that were performed.

The results were consistent across:
Model addressing and have performed validity checks for;
Alternatively, varying variables were used.

5. ANOVA and ANCOVA as Regional Analysis Techniques

5.1 Rationale

Examination of the data was performed through exploratory analysis and indicated there are clear geographic patterns of health outcome data, specifically, high levels of obesity, hypertension, and diabetes among those from the southern region of the United States. However, it is possible that the differences in health outcomes across geographic regions stem more from the underlying economic structure, rather than the geographic regions themselves.

To statistically assess if this assumption is true, we will conduct the following analyses:

- ANOVA → to assess if mean differences exist among the various geographic regions.
- ANCOVA → to assess if these mean differences exist after controlling for the underlying economic variables associated with each region.

This analysis is critical to help address the following:
- ANOVA assesses if the various regions differ.
- ANCOVA assesses if the various regions still differ after accounting for the economic structure of each region.

5.2 One-Way ANOVA

We will begin our analysis with a one-way ANOVA analysis.

$obesity=\mu+\alpha\_region+\epsilon$
Findings:
Strong evidence of statistically significant between-region differences in obesity levels.

The mean level of obesity among census regions differed significantly based on region:

The highest obesity prevalence in the southern region of the United States.

The lowest prevalence of obesity is in the western or northeastern United States.

5.3 Shortcomings of the ANOVA Method

The one-way ANOVA test confirms statistically significant differences based on region (e.g., south versus west/northeast) related to the level of obesity; however, it does not identify the reason(s) for these differences (e.g., based on:

- Differences in the economic profile of the regions.
- Differences in the structural make-up of the regions.
- Differences in the institutional arrangements in the regions.
-

Therefore, the analysis using only one-way ANOVA cannot determine if the overall differences based on region (e.g., obesity) are a result of economic conditions or due to deeper structural issues (e.g., lack of access to healthcare).

5.4 ANCOVA: Control for Economic Factors

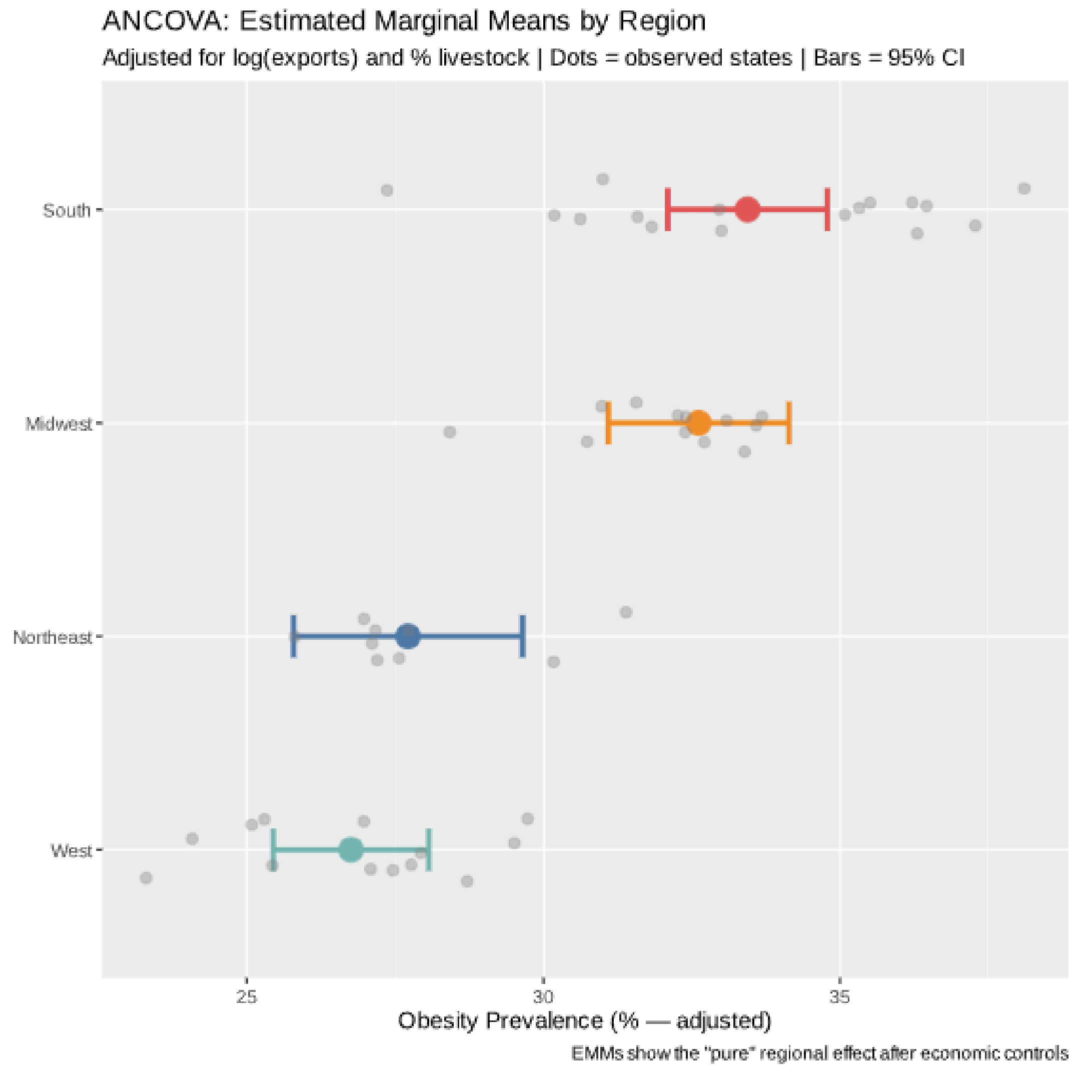
The following model allows for estimation of the relationship of,

$$obesity=\beta0+\alpha\_region+\beta1log(exports)+\beta2pct\_livestock+\epsilon$$

To control for regional comparisons, we are adjusting for

- economic comparative scale
- economic structural factors

5.5 ANCOVA Findings



The results of the ANCOVA indicate that

- Region remains statistically significant
- pct_livestock no longer statistically significant with region included
- log(exports) shows very weak evidence

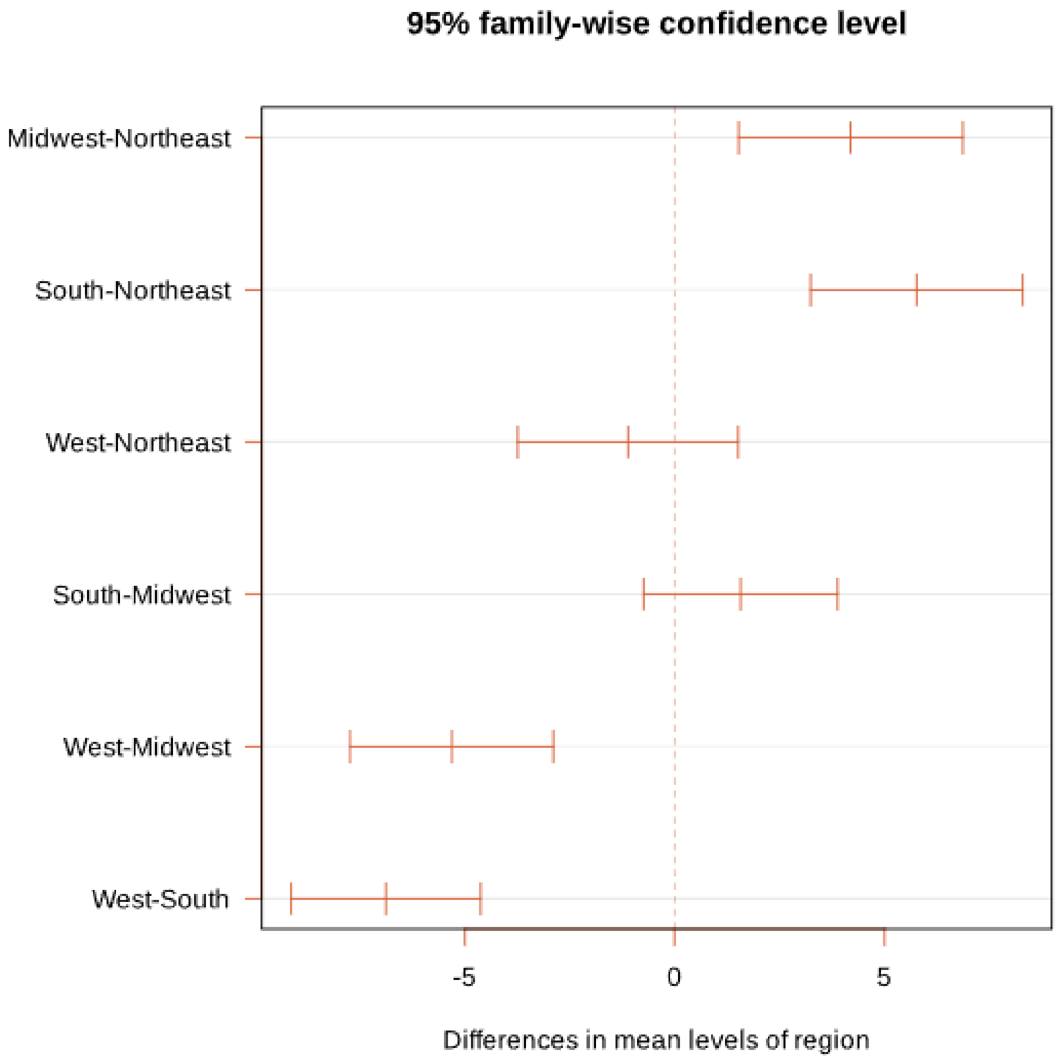
5.6 Interpretation

This finding is critical for multiple reasons...

5.6.1 Persistence of Regional Effects

```
=== ONE-WAY ANOVA: obesity ~ region ===
            Df Sum Sq Mean Sq F value   Pr(>F)    
region       3  431.2  143.74   27.87 2.01e-10 ***
Residuals   46  237.2    5.16                     
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Eta-squared (η²): 0.645
→ LARGE effect (η²>0.14)

=== LEVENE'S TEST (homogeneity of variances) ===
Levene's Test for Homogeneity of Variance (center = median)
      Df F value  Pr(>F)  
group  3  4.1711 0.01077 *
      46                  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
→ Unequal variances. Use Welch ANOVA (oneway.test).

=== TUKEY HSD POST-HOC COMPARISONS ===
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = obesity_pct ~ region, data = df)

$region
                       diff        lwr       upr     p adj
Midwest-Northeast  4.197222  1.5279485  6.866496 0.0006983
South-Northeast    5.770139  3.2479123  8.292365 0.0000012
West-Northeast    -1.111111 -3.7360171  1.513795 0.6741511
South-Midwest      1.572917 -0.7387422  3.884576 0.2801240
West-Midwest      -5.308333 -7.7316103 -2.885056 0.0000030
West-South        -6.881250 -9.1415323 -4.620968 0.0000000
```

```
West-South         -6.881250 -9.141525 -4.620968 0.0000000
```



Despite the presence of economic factors, regional differences still exist.
This indicates that there are factors that explain geographic disparity in economic structure that are unrelated to geography, additional structural and/or institutional factors

5.6.2 Rationale for Loss of Statistical Significance for pct_livestock

This is not a discrediting of this variable; rather, an insight.
Possible rationale:
Economies that have a heavy reliance on livestock products tend to be concentrated geographically in Southern regions. When accounting for the effect of region, part of the variance in pct_livestock and geographic areas is accounted for.
In conclusion, Economic structure and geography have some overlap in the explanation of obesity in the USA.

5.6.3 Implication

The health disadvantage in the southern region hinges on more than just economic factors. It is also likely affected by the following (in no particular order):

- Differences in access to healthcare
- Variances in policy environment
- Long-term socioeconomic inequality
- Differences in patterns of culture and behavior

5.7 Pairwise Comparisons

Post-hoc analysis through (Holm-adjusted) tests shows that:
The southern region differs significantly from other regions.
Non-southern regions have much weaker differences between them compared to southern and non-southern regions.
Interpretation
The southern region and the other regions of the United States represent the most significant divide in the country.

5.8 Interaction Effects

We evaluated the effects of obesity by:

Obesity = (region, percentage of the livestock to total population, log of exports)
Result
The interaction term did not improve the fitting of the model (as evaluated by AIC)
Interpretation
Generalizes to all regions regarding the share of livestock on obesity.

5.9 Synthesis of Regional Analysis

This section synthesizes previous findings and extends previous results by demonstrating:

- Regional difference is not only real but substantial, as indicated by ANOVA
- Economic factors alone do not fully explain regional variation, as indicated by ANCOVA
- The economic structure/ geography is partially confounded, as shopping for local products does not influence livestock share once the region is accounted for

- There is no strong interaction, as indicated by the fact that the economic structure has the same relationship to health by region.

6. Nonlinear Modeling, Classification, and Regularization

This section examines how the results of linear regression are sensitive to the assumptions of the model, dependent on the functional form of the model, and affected by overfitting. This will be done using three different methods:

Generalized Additive Models (GAM) will be used to determine whether or not non-linear relationships exist in the data; logistic regression will be used to derive a probabilistic interpretation of the variables; LASSO and ridge regression will be used for regularization and selection of variables as well as examining robustness.

6.1 Generalized Additive Model (GAM)

6.1.1 Rationale

Linear regression assumes a straight-line relationship among all the predictors and the outcome. However, actual relationships may have:

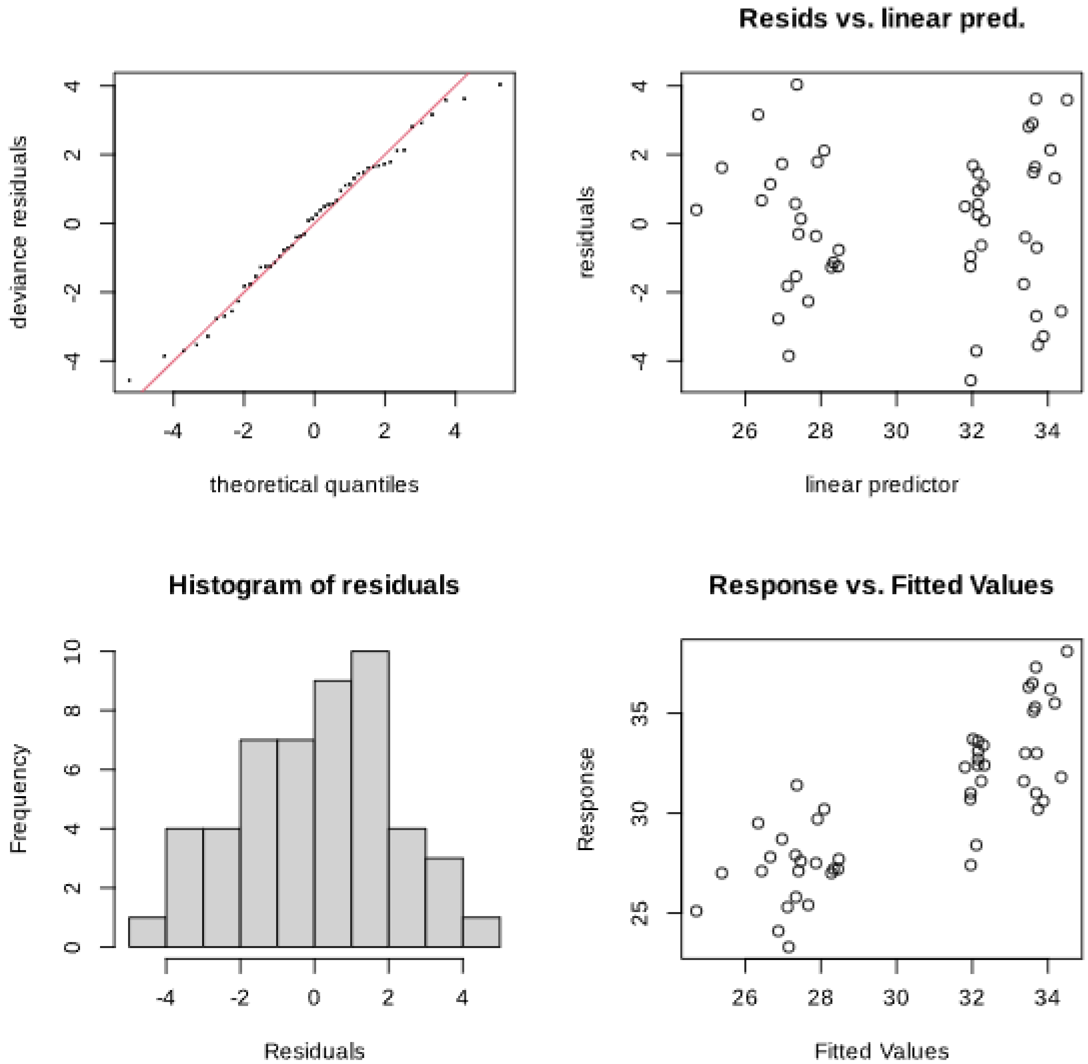
- Thresholds
- Diminishing returns
- Non-linear curvature

To test for this, a GAM will be estimated, providing for flexible smooth functions; for instance:

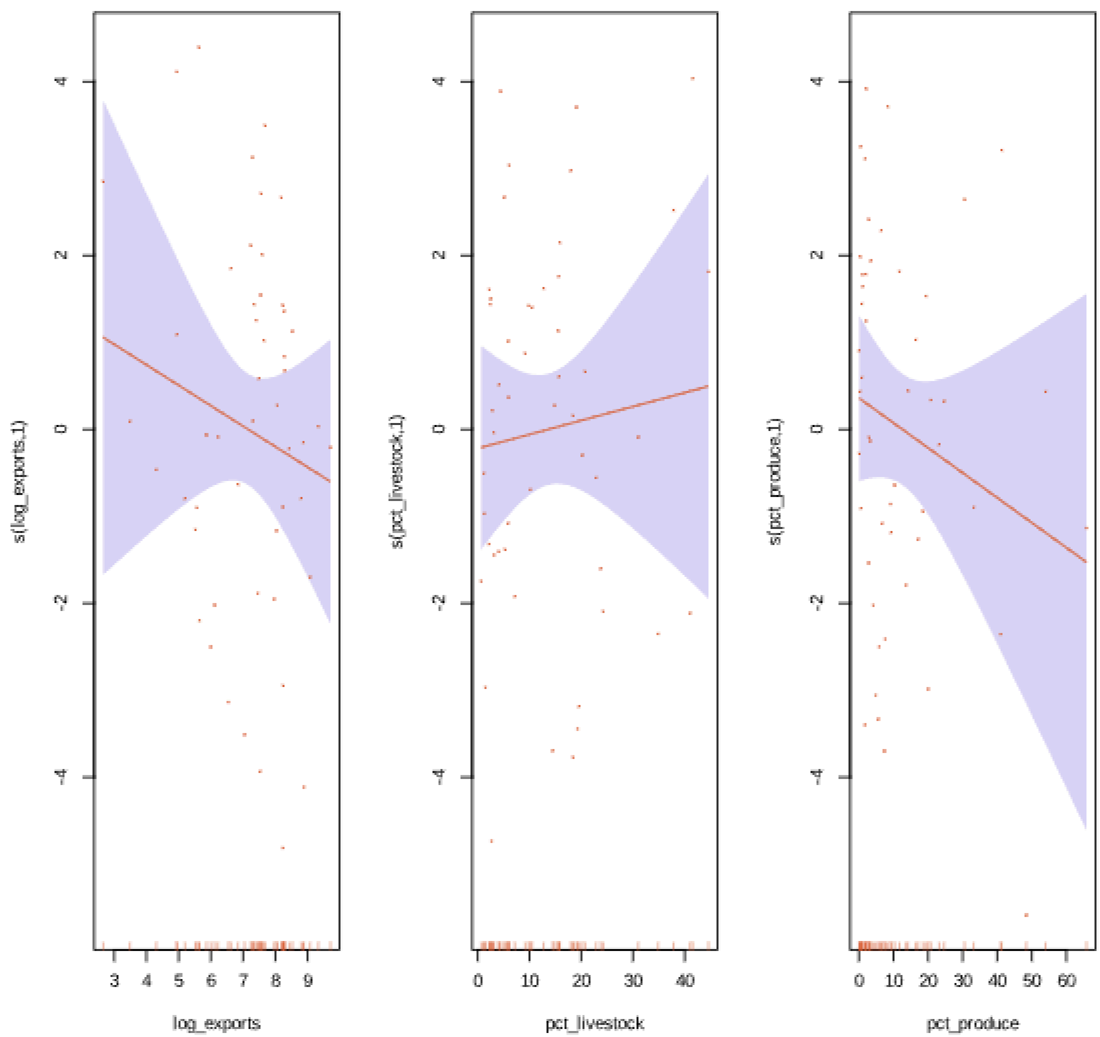
- obesity=s(log(exports))+s(pct_livestock)+region+e

6.1.2 Major Findings

- The effective degrees of freedom (edf) associated with the smooth terms are approximately one.
- There was not strong evidence of curvature in the data.
- The GAM's fit is similar to the OLS fit (i.e., comparable AICs).




```
=== READING THE SMOOTH PLOTS ===
Y-axis: partial effect of this predictor (centered at 0)
Shaded band: 95% confidence interval
Rug marks at bottom: observed data values
A near-straight line (edf≈1) confirms OLS linearity assumption.
A curved line reveals threshold effects OLS would miss.
```



6.1.3 Interpretation (of the GAM)

GAM results indicate that relationships between predictor variables and obesity are almost all linear in nature, which implies that:

- Linear models are sufficient (there is minimal benefit in applying a more flexible non-linear approach).
- The results are not a function of model misspecification; therefore, main findings (i.e. livestock effect, regional effect) are robust to relaxing linearity assumptions.

6.1.4 Model Diagnostics

- Basis dimension (k) is adequate (k-index is greater than one).

- There was low concurvity as it pertains to non-linear multicollinearity
- Residual diagnostics showed no major violations.

6.1.5 GAM Conclusion

Data do not support the presence of complex non-linear relationships; therefore, making simpler, interpretable models significantly more reliable.

6.2 Logistic Regression

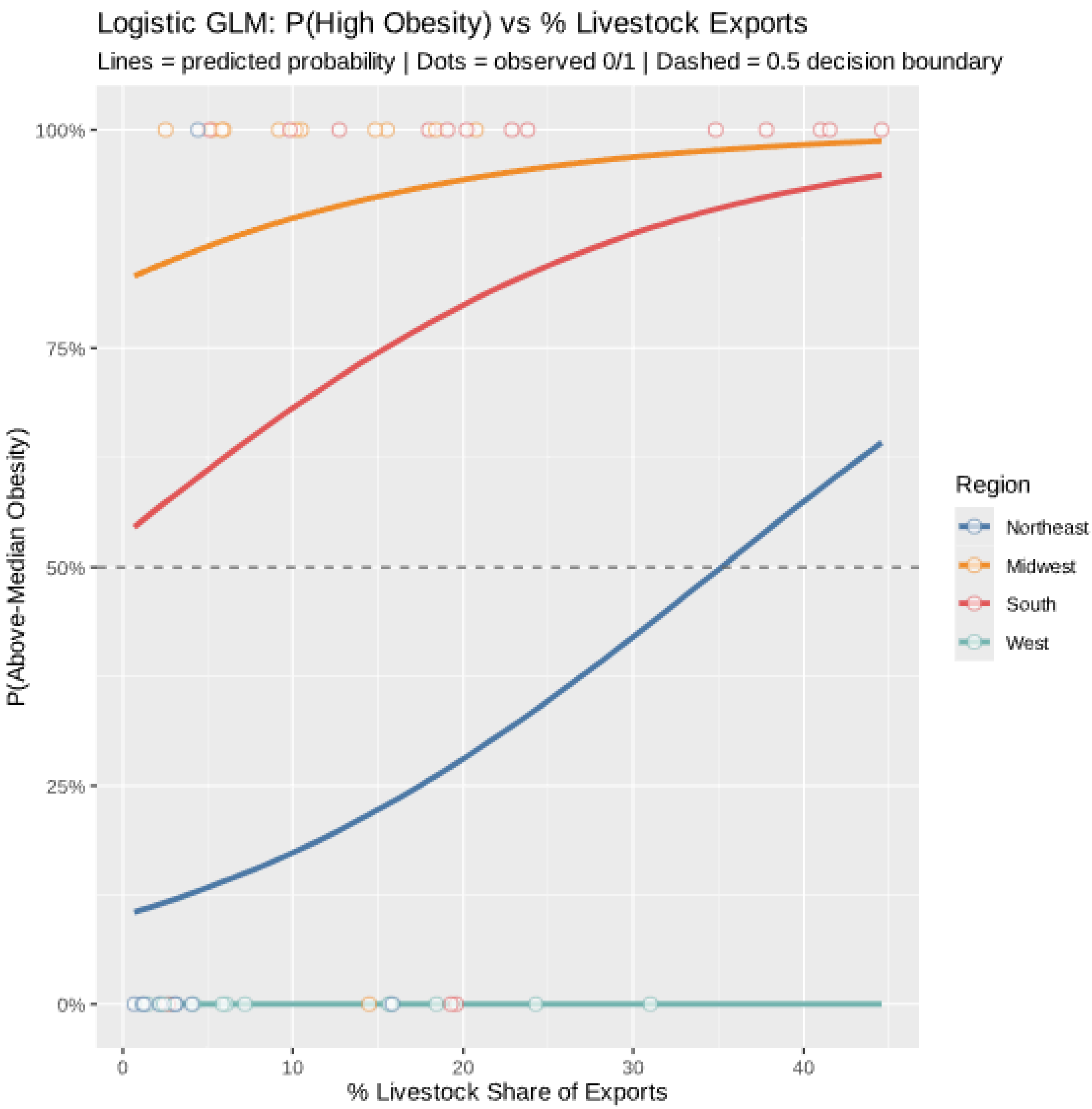
6.2.1 Rationale

The OLS (ordinary least squares) model estimates the continuous nature of the prevalence of obesity, but policymakers are usually concerned with classifying risk (e.g., which states are at the highest risk for elevated obesity). To meet this need, we define a binary outcome variable:
high_obesity=1 if the obesity level is greater than the median.
Substituting for high-obesity (HOB) with generalised ordered logit models (GOLMs), we can restate the GOLM as follows:

6.2.1 Model Specification

$P(HOB) = \exp(\beta 0 + \beta 1 \log(exports) + \beta 2 pct\_livestock + \beta 3 pct\_produce + \gamma region)$

6.2.2 Results



pct_livestock remains the strongest positive predictor of HOB, regional effects remain significantly associated with HOB, and log(exports) remains a weak predictor of HOB.

6.2.3 Classification Performance

Accuracy for classifying HOB states > 80%, threshold = 0.5
Important Concept: In interpreting classification accuracy, it is important to keep in mind the baseline prevalence of HOB.

6.2.4 Interpretation

States with a relatively large share of livestock are relatively more likely to be classified as HOB states, consistent with previous findings from regression analyses; therefore, this study significantly improved on previous findings using a different statistical approach and hence, provides several new insights regarding the relative risks of HOB among those states.

6.2.5 Limitation

The selection of a threshold for classification at 0.5 is purely arbitrary, and the GOLM model does not include an explicit adjustment for class imbalance.

6.3 Regularization: LASSO and Ridge

6.3.1 Motivation

The presence of relatively few (n = 50) observations relative to the number of predictors has created the potential for the model parameters to be either overfitted or unstable. In order to alleviate this problem, regularization has been proposed as an approach to penalizing complexity in models.

6.3.2 LASSO (Variable Selection)

LASSO minimizes:

$RSS+\lambda\sum|\beta j|$
- Key Result = pct_livestock retained, weaker variables were shrunk towards zero
- Interpretation = LASSO identifies pct_livestock as the most important predictor.

6.3.3 Ridge Regression (Stability)

Ridge minimizes:
$RSS+\lambda\sum\beta j^2$
- Key Result = the shrinkage of coefficients while maintaining their stability; there are no variables that were totally removed.
- Interpretation = Ridge has resulted in enhancement of stability, but not in alteration of the underlying relationships.

6.3.4 Cross-Validation

- The optimal lambda was found using 10-fold CV
- The CV provides an unbiased estimate of the prediction error

6.3.5 Model Comparison

- OLS = lowest in-sample (biased)
- LASSO/Ridge = provides better estimates out of sample.

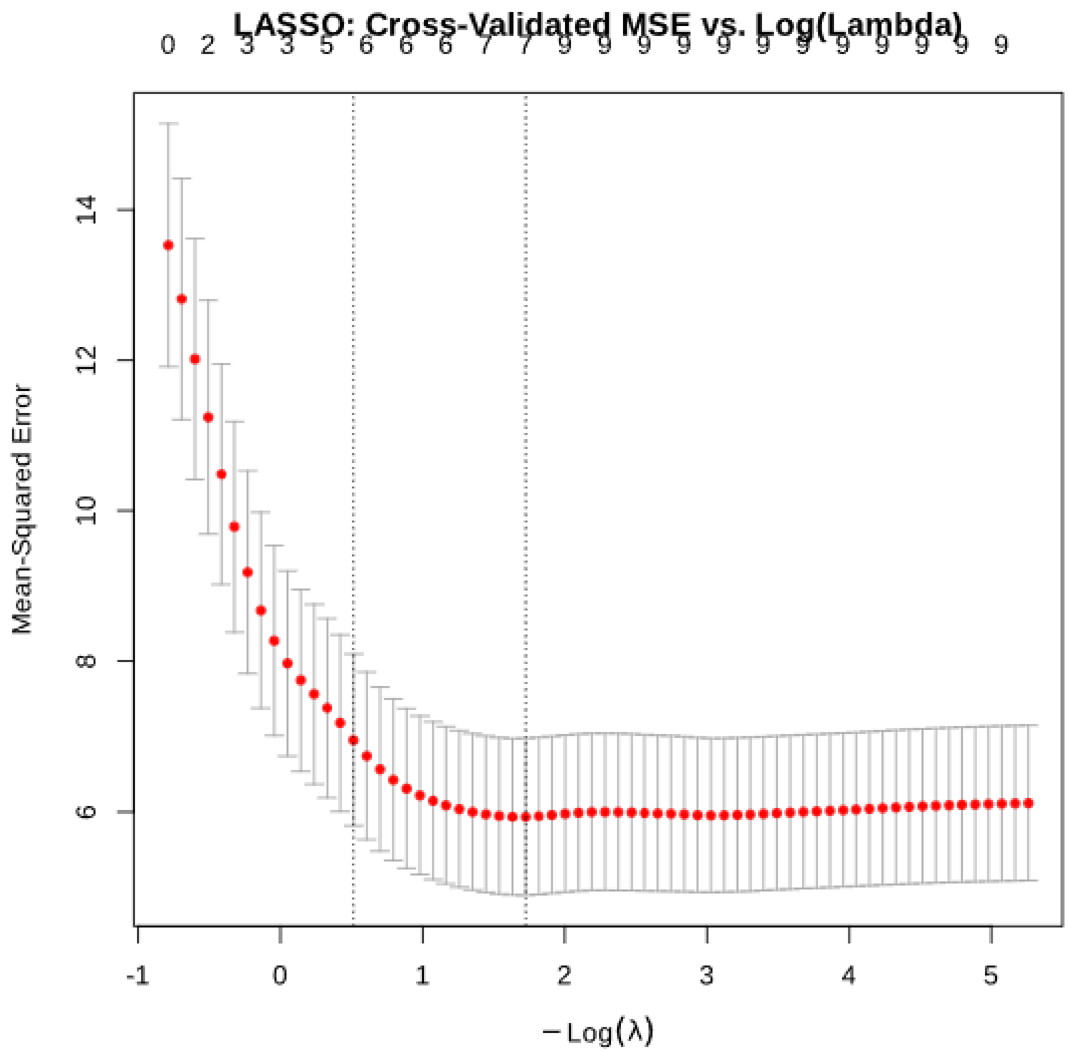
6.3.6 Synthesis

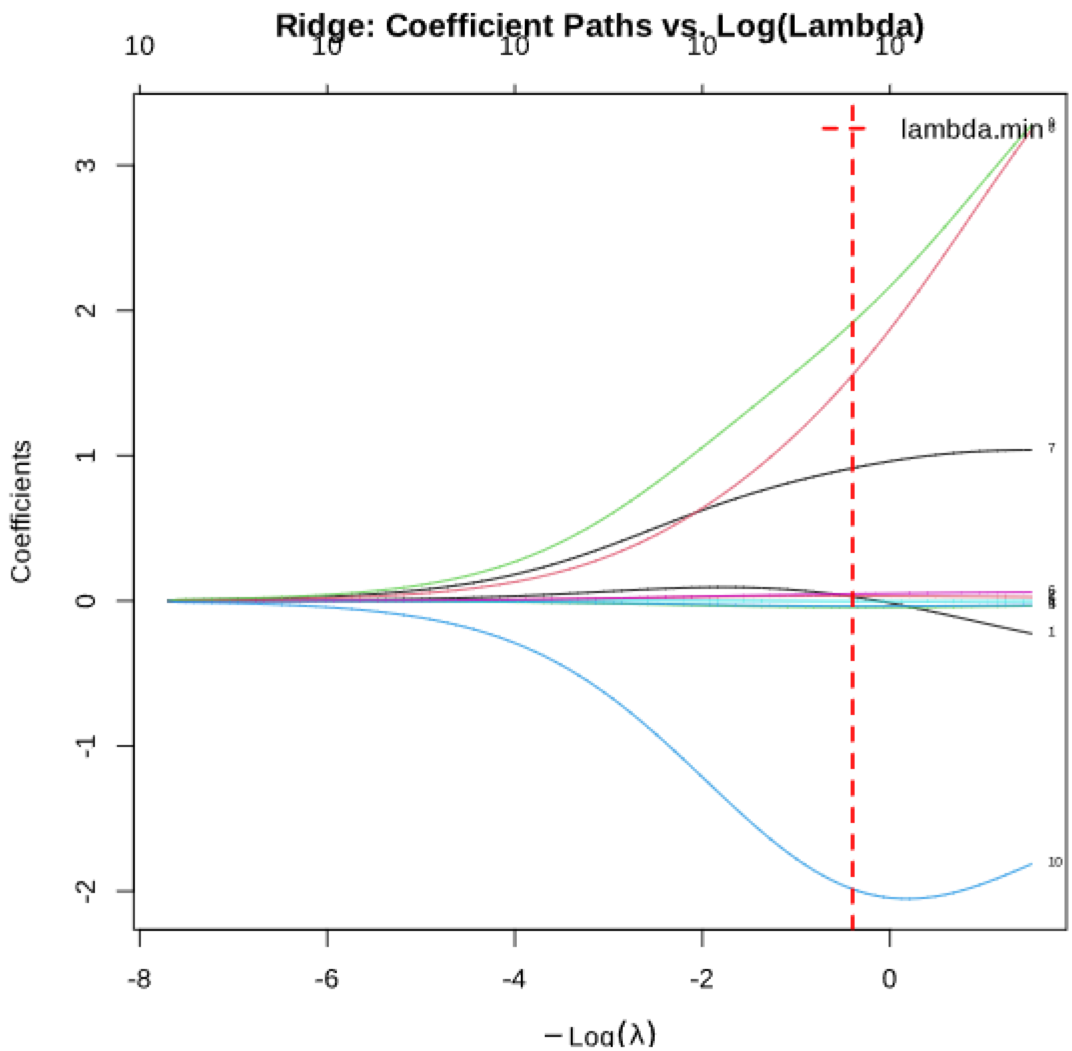
- Regularization verifies that:

The results have not been due to overfitting, as the key predictors are stable and the model can be generalized to a greater population than that sampled.

6.4 Integrated Interpretation

```
→ LASSO zeroes out weak predictors; retained predictors are most robust
```





1. Livestock share has remained the most dominant predictor among all of the advanced methods.
Livestock share appears consistently in:
- OLS
- GAM
- Logistic Regression
- LASSO

2. Strong regional impact
    - Could not be eliminated through:
    - structurally controlled
    - nonlinear modelled
    - regularised.

3. Linearity of assumption has been validated
    - concurred by GAM -

- no strong non-linear patterns
- that OLS modelling is appropriate.

4. Consistency of results
Results have been consistent across:
- Functional forms
- Outcomes
- Estimating Procedures.

7. Conclusion and Implications

The purpose of the study was to investigate the relationship between health outcome differences (such as asthma) and varying agricultural economic structures across states in the United States of America. Utilizing a comprehensive analysis that was conducted using a variety of modeling techniques (including linear regression, ANOVA/ANCOVA, generalized additive models, logistic regression and regularization), several key conclusions emerged.

1. Economic Structure is More Important Than Economic Size

Using log-transformed exports as a proxy measure of total agricultural output of a state has shown weak and inconsistent relationships to health outcomes; conversely, using economic structure, in the form of production composition, specifically the proportion of livestock raised in the state, the composition of agricultural output regarding livestock has a statistically significant and stable predictive value for obesity and hypertension rates. This indicates that the product produced by a state provides a better explanation of the variation of population health outcomes than does the quantity that is produced.

2. The Proportion of Livestock Raised in a State is the Most Valid Predictor

Across all modeling approaches, the percentage of livestock raised in a state showed positive correlations to the rate of obesity and hypertension.

Through the analysis of multiple modeling methods, the evidence suggests that the relationship between the percentage of livestock and these health outcomes is not solely an artifact of the chosen statistical model configuration, but rather represents a true empirical relationship.

3. Regional Disparities Exist Beyond the Scope of Finance

Even when accounting for:
• Economic Scale
• Economic Structure
• Economic Diversification

The regional incidence (specifically the large health disparity in the South) is still found to have statistically meaningful results.

This evidence indicates that the geographic disparities observed demonstrate underlying structural, institutional, or historical issues beyond those that can be measured through economic indicators.

4. The Relationships Between Variables Are Generally Linear

The GAM shows that:
• Effective df's = 1
• No strong nonlinear characteristics

Thus, the results may be modelled with linear modelling and could produce strong interpretability.

All Findings Remain Strong Across Methodologies.

The core findings hold true despite:
- The functional form used (linear or non-linear)
- The type of dependent variable (continuous or binary)
- The estimation approach used (OLS or generalised estimating equations)

The cross-method consistency aids in increasing confidence in the conclusions.

7.2 Interpretation and Broader Insights

Based on these findings, we conclude:

Agricultural economic structure is a significant yet underrated factor in population health outcomes.

At the same time, we recognize that the results reveal an important limitation of a solely economic approach to explaining outcomes.

While economic factors help explain some of the differences we observe, they do not explain all of them.

The midwestern dominance of both the highest and lowest population health outcomes is a clear illustration of regional effects continuing to exist, after adjusting for economic variables.

7.3 Limitations

Despite being methodologically sound, there are limitations to this analysis.

1. Cross-sectional Nature
As the data are collected from a single point in time (2019), they do not allow us to infer:
Attempts to assess temporal relationships and changes of a dynamic nature over time

2. Ecological Analysis
All data sets are aggregated at the state level.
Therefore:
The results do not represent the relationship of variables at an individual level (e.g., individuals living in the same state do not necessarily experience the same health outcome).
Ecological fallacies may occur from making assumptions about how individuals would behave based on the results of the study aggregated at the state level.

3. Omitted Variable Bias
There may be omitted variable bias with several omitted variables (e.g., healthcare access, income inequality, education, policy, et cetera) that could confound the results.

4. Small Number of Observations

Using 50 subjects yields:
Statistics have less power
The results can change considerably with small alterations
Regularisation techniques sometimes help relieve this issue.

5. Limitations in Measurement of Data

Export data represents goods produced while export medical data is reported by consumers (BRFSS).

7.4 Conclusion: Research and Policy Implications from Limitations

Some of these results have considerable relevance despite the previously mentioned limitations.
(1) Economic structure is a significant factor in Public Health research.

Current methodology presently only assesses the impact of income as a contributing factor to health; when an understanding of economic structure is realised would yield additional information about the determinants of health.
2. Disparities between Regions Need More Than Economic Response

Regional Disparities Will Generally Still Exist Once An Economy Has Adjusted

To deal with the effects of regional disparities due to the adjusted economy:

Structural Inequalities Need To Be Addressed By Policy Responses

Economic Interventions Alone Will Not Be Enough To Solve The Problems Of Regional Disparity

3. Agricultural Policy May Impact Public Health Indirectly

If The Economic Structure Affects The Health Outcomes Then Agricultural Subsidies, Production Incentives, And Trade Policy Will Have An Impact On Public Health.

7.5 Future Directions for Research

This Study Provides a Basis for Future Research.

The Future Of The Study Can Look To Extend The Current Analysis By:

1. Using Longitudinal Data To Track The Changes Of Time And To Identify Dynamic Relationships

2. Possibly Looking At More Specific Geographic Areas By County Or Individual Level to Reduce The Aggregation Error

3. Using Causal Inference Methods Such As Instrument Variables, Natural Experiments, And Policy Shocks To Assess The Effects Of Policy Changes

4. Addition of Both Environmental Influences and Individual Factors as Potentially Associated Variables
Access to healthcare services
Socioeconomic determinants
Behavioral risk characteristics

5. Potential underlying pathways
Nutritional habits
The environment where one lives or works
Working environment

Conclusion:
Based on the results of this study, the following conclusions can be made:

A state's agricultural economic system (particularly proportions of animal agriculture) is related to variation in population-level health results among states in the USA.

At the same time, regional differences exist beyond what can be explained by the economy, indicating that there are structural and institutional factors that contribute to regional differences.

The ability of different statistical analyses to yield the same set of results demonstrates that the relationships are strong and do not depend on a single assumption made about the data.

Therefore, these findings lead to a better understanding of how economic systems shape health, and suggest the need for further research into causality.